

MODULE – V

Traditional Knowledge in Professional domain

Town planning and architecture

It reflect deep-rooted principles of sustainability, community, spirituality, and harmony with nature. These systems, often overlooked in modern urban development, offer valuable insights into creating livable, resilient, and culturally meaningful settlements.

1. Arthashastra in Town Planning

Arthashastra, attributed to **Kautilya (Chanakya)** around the 4th century BCE, is a foundational Indian treatise on statecraft, economics, and military strategy. It also offers one of the earliest documented frameworks for **urban planning and architecture** in ancient India. The town planning ideas in the *Arthashastra* are remarkably systematic, functional, and ahead of their time.

Aspects of Town Planning in the Arthashastra

1. Site Selection

- Towns should be located near natural resources (water, fertile land, forests).
- Emphasis on strategic defensibility and trade routes.
- Consideration of climatic conditions and disease avoidance.

City Structure and Layout

- **Fortification:** Cities were to be **fortified with ramparts, moats, and gates**. The fort (durga) was the nucleus of the city.
- **Rectangular Grid System:** The Arthashastra recommends **a regular layout**, often interpreted as a **grid-based plan** — straight roads intersecting at right angles.
- **Zoning:**
 - **Residential zones** for different castes and professions.
 - **Market areas** placed centrally for accessibility.
 - **Administrative zones** for state offices, treasuries, and the palace.
 - **Military zones** for barracks, weapon stores, and guard towers.

3. Infrastructure and Utilities

- **Roads:** Well-planned wide roads and smaller lanes; importance of cleanliness and maintenance emphasized.
- **Water Management:**
 - Construction of **wells, canals, reservoirs, and drainage systems**.
 - Rainwater harvesting encouraged.
- **Waste Management:** Arthashastra prescribes designated areas for waste disposal and strict penalties for public littering or pollution.

4. Safety and Surveillance

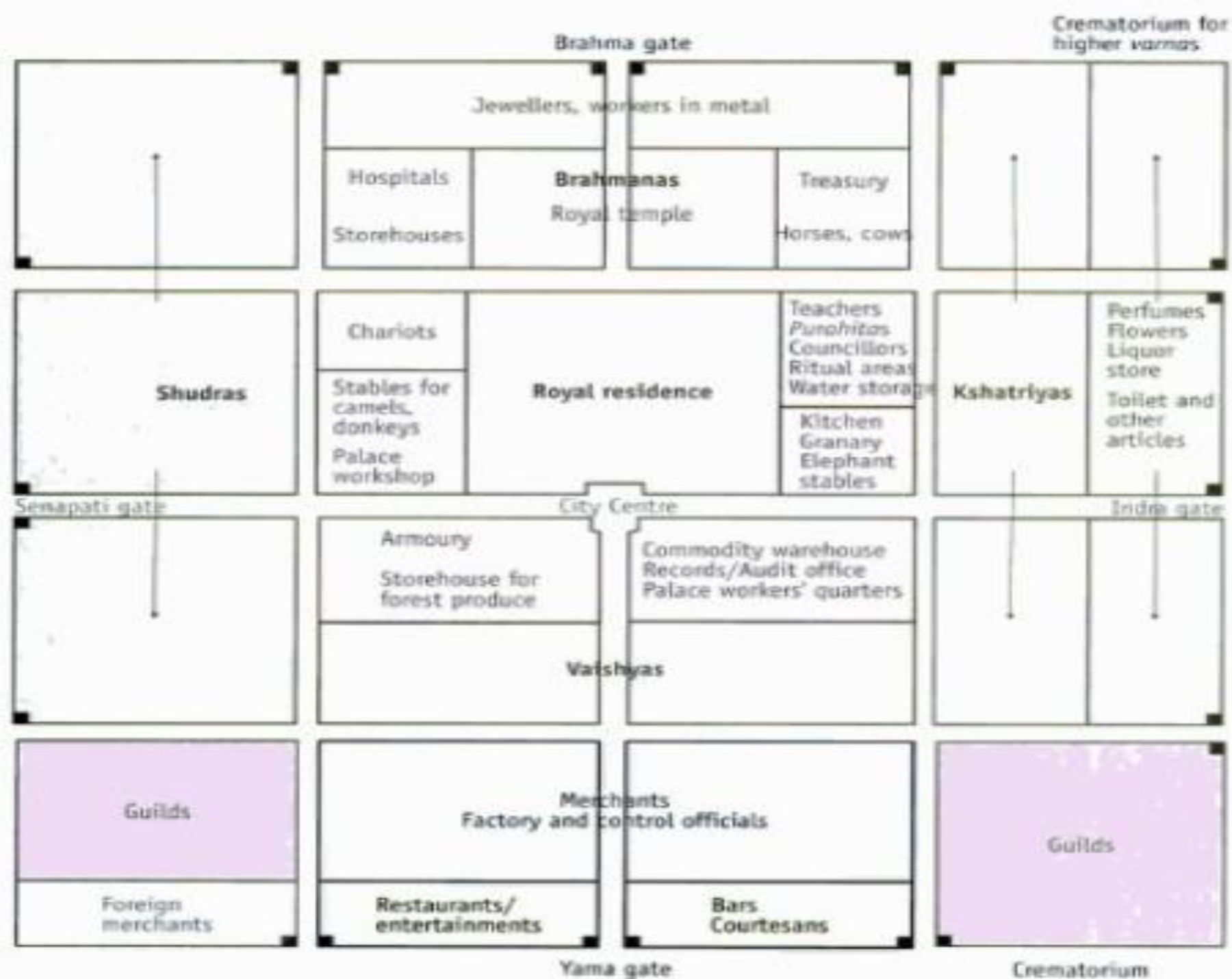
- **Watchtowers and secret agents** to monitor crime, corruption, or rebellion.
- Residential registers and movement tracking within the town.

5. Public Amenities

- **Hospitals, schools, temples, and gardens** were recommended as part of public planning.
- **Markets** regulated for fair trade, weights and measures.

6. Disaster Preparedness

- The city should be prepared for **fire, floods, enemy attack**, and famine — including maintaining emergency stores of grain and arms.



Raja Dharma (Arthashastra)

Raja Dharma in the Artha Shastra refers to the duties and responsibilities of a king as described by Kautilya (also known as Chanakya or Vishnugupta). The *Artha Shastra*, written around the 4th century BCE, is a foundational text of political science and governance in ancient India. "Raja Dharma" specifically focuses on the ethical and administrative obligations of the ruler in maintaining order, prosperity, and justice.

1. Raksha
2. Vyavahara
3. Palana
4. Kosha
5. Danda Niti
6. Espionage and Diplomacy
7. Advisor and Ministerial Governance
8. Moral Conduct and Self-Discipline

Comparison of Raja Dharma in Arthashastra vs Manusmriti

Aspect	Artha Shastra (Kautilya)	Manusmriti (Manu)
Nature of the Text	Secular, pragmatic treatise on politics and economics	Religious-legal code rooted in Dharma (moral law)
Focus of Raja Dharma	Practical governance, administration, and statecraft	Upholding Dharma, caste duties, and moral justice
Role of the King	Strategic ruler focused on law, order, prosperity, and power	Moral and spiritual guardian enforcing divine law
Justice (Vyavahara)	Justice serves state stability; based on evidence and reason	Justice must align with Dharma Shastras and caste laws
Use of Punishment (Danda Niti)	Central to maintaining order and discipline; scientifically applied	A divine tool to protect righteousness and deter sin
Welfare of the People	Economic development, fair taxation, infrastructure	Ensure moral order and caste-based duties
Spying and Espionage	Encouraged as a key tool of internal and external control	Not emphasized; considered morally questionable
Military and Diplomacy	Realpolitik: war, alliances, espionage based on state interest	Emphasis on righteous war (Dharma Yuddha)
Ministers and Counsel	Governance through trained advisors and experts	Advisers should be Brahmins; spiritual wisdom valued
Religious Role of King	Minimal; religion is secondary to political power	Strong; king must uphold Vedic rituals and Brahmin supremacy
Taxation and Revenue	Taxes based on productivity; economy-centric	Taxes should maintain Brahmins and societal order

Kautilya Arthashastra

The **administrative setup in Kautilya's Artha Shastra** is one of the most detailed and sophisticated models of governance found in ancient Indian literature. It outlines a **centralized, hierarchical bureaucracy** designed to ensure efficient governance, control, and economic prosperity.

1. Raja
2. Mantri Parishad
3. Amatyas
4. Tirthas
5. Provincial Administration
6. District and Village Administration
7. Gudhapurusha
8. Justice System
9. Revenue and Taxation

Health and Disease treatment on Unani

Unani is based on **four bodily humors** (*Akhlat*), and health is defined as the **balance** of these:

1.Dam (Blood) – Hot & Moist

2.Balgham (Phlegm) – Cold & Moist

3.Safra (Yellow Bile) – Hot & Dry

4.Sauda (Black Bile) – Cold & Dry

- Each person has a **temperament (Mizaj)** influenced by these humors, and treatment aims to **correct imbalances**.

1. Tahaffuz-e-Sehat
2. Ilaj
 - a. Ilaj bil Tadbeer
 - b. Ilaj bil Ghiza
 - c. Ilaj bil Dawa
 - d. Ilaj bil Yad
3. Humma

The **Unani system** maintains health through **lifestyle regulation and natural balance**, and treats disease by **restoring humoral equilibrium** using:

- **Preventive care (Six essentials)**
- **Natural treatments (diet, regimen, drugs)**
- **Emphasis on individual temperament and balance**

Indian Epics- Public Administration

1. Ramayana

2. Mahabharata

Theme

Dharma (Ethical Duty)

Rajadharma (Duty of Kings)

People-Centric Governance

Advisor System

Decentralization

Accountability

Description

Rulers must uphold righteousness above personal gain.

Protect people, punish the wicked, and ensure justice and prosperity.

Welfare of citizens is central; rulers are **servants of the people**.

Emphasis on wise counsel (e.g., Vasishtha, Vidura, Bhishma).

Local administration and village-level self-governance are encouraged.

Kings are answerable to divine law, sages, and public conscience.

Ancient India- Treasury and State Economy

In **Ancient India**, the **treasury (Kosha)** and **state economy** were crucial elements of governance, especially as described in texts like the **Artha Shastra**, **Manusmriti**, and the **Indian epics**. The system was well-organized, revenue-driven, and linked closely with agriculture, trade, and ethical governance.

1. The Treasury (Kosha)

- The **Kosha** was the **financial backbone** of the kingdom.
- It was **maintained by the king**, advised by the **Koshaadhyaksha** (Superintendent of Treasury).
- Funds were used for:
 - Military expenses
 - Infrastructure (roads, irrigation)
 - Public welfare
 - Salaries of officials
 - Diplomatic missions and festivals

Source

Land Revenue (Bhaga)

Trade and Commerce Taxes

Mines and Forests

Tributes and Fines

Cattle and Animal Wealth

State Monopolies

Description

Main source; share of agricultural produce, usually 1/6th.

Customs duties, tolls, and merchant taxes.

Revenue from minerals, salt, and timber.

Payments from vassal states and judicial fines.

Taxes on livestock and dairy.

Salt, alcohol, metals, and arms production.

State Economy

- The economy was **agrarian**, but also supported by **trade, crafts, and industry**.
- Emphasis on **self-sufficiency**, local production, and **ethical trade**.

Sector

Features

Agriculture

State invested in irrigation, seeds, and land reforms.

Industry and Crafts

State promoted weavers, blacksmiths, potters through guilds (Shrenis).

Trade (Vanijya)

Both inland and maritime trade flourished. Ports like Lothal and Bharuch were vital.

Banking and Loans

Moneylenders existed; interest rates were regulated by law.

Coinage

Coins (e.g., punch-marked coins) made of silver, gold, and copper were in circulation.

Management and Accountability

- **Auditors** ensured that state funds were not misused.
- **Regular inspections** of treasuries, markets, and granaries.
- **Punishment for embezzlement** was severe under Kautilya's system.
- Surplus was **stored in emergency reserves** for famines or wars.

Philosophy of Economy

- Economy served the **Dharma of the ruler**: welfare of subjects.
- A prosperous state was seen as one where **people are productive, taxes are fair**, and the treasury is full.

Siddha System for Treating Human Diseases

Siddha medicine is based on the belief that **the body is composed of the five elements (Panchabootham):**

1. Earth (Prithvi)
2. Water (Appu)
3. Fire (Thee)
4. Air (Vayu)
5. Space (Akasam)

These elements combine to form **three humors (Mukkuttram)** that govern bodily functions:

- **Vatham** (Air + Space): Controls movement.
- **Pitham** (Fire): Governs metabolism and digestion.
- **Kapham** (Earth + Water): Maintains stability and structure.
- **Health = Balance of Mukkuttram**
Disease = Imbalance of Mukkuttram

Diagnosis (Noigal Neralvu Murai)

- Diagnosis is both **physical and spiritual** and includes:
 - **Pulse reading (Nadi Pariksha)**
 - Observation of tongue, eyes, urine, stool, voice, and skin
 - Evaluation of patient's lifestyle and temperament (Udal Kattugal)

Treatment Principles

1. Unavu Maruthuvam
2. Mooligai Maruthuvam
3. Thathu Maruthuvam
4. Yoga and Kayakalpa
5. Sirappu Maruthuvam
6. Varmam and Uzhichil
7. Parpam, Chendooram
8. Dinacharya and seasonal routine Ritucharya

Kautilya's Artha Shastra-Saptaanga

No.	Element (Sanskrit)	Meaning / Role in Governance
1.	Swamin (The King/Ruler)	The head of the state who governs according to Dharma; responsible for security, justice, and welfare.
2.	Amatya (Ministers/Officials)	Well-trained advisors and administrators who assist in governance and policy execution.
3.	Janapada (Territory & People)	The land and its population; includes citizens' welfare, productivity, and protection.
4.	Durga (Fortified Capital)	The administrative and military center; emphasizes town planning, defense, and infrastructure.
5.	Kosha (Treasury)	State finances; supports administration, welfare programs, defense, and economic development.
6.	Danda (Law and Force)	The system of justice and law enforcement; ensures order through fair punishment and discipline.
7.	Mitra (Ally)	Foreign relations and diplomacy; allies support defense, trade, and political stability.